Choosing a carryover-mitigation analysis strategy for biomarker-treatment interactions in aggregated N-of-1 trials: a simulation-based tutorial and factorial evaluation

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Abstract

Background. Aggregated N-of-1 trials – in which each participant is repeatedly crossed between treatment and control and many such single-patient series are pooled – are an increasingly used design for validating predictive biomarkers, but they are unfamiliar to many applied statisticians. A recurring inferential target is the biomarker-treatment interaction, and a recurring nuisance is carry-over: residual drug effect that decays after discontinuation and contaminates off-drug observations. The analyst must decide how to represent residual exposure in the linear predictor, yet the joint sensitivity of the interaction test to mis-specification at both the data-generating and analyst layers has not been characterized. This paper is written as a tutorial: we introduce the design and the modeling choices from first principles, then evaluate them.

Methods. We compare three analysis-model specifications for the drug-exposure regressor: a binary on-drug indicator that ignores carryover (A1); an exposure-weighted continuous indicator that assumes a parametric decay and half-life (A2, the current default); and the binary indicator augmented by a lagged just-off-drug term (A3, the classical crossover device). We embed them in an ADEMP-structured factorial simulation[1] crossing three data-generating process (DGP) decay forms (linear, exponential, Weibull) with the three specifications, under two biomarker data-generating architectures (mean moderation and multivariate-normal (MVN) differential correlation), three designs, two sample sizes, and three interaction strengths (540 cells, 500 replicates each), plus single-axis sensitivity blocks and a cluster-robust recalibration. Every performance measure is reported with its Monte Carlo standard error (MCSE).

Results. The exposure-weighted specification A2 does not dominate. It attains the highest power only under the differential- correlation architecture (B) and only in the two designs with a blinded-discontinuation phase, where its advantage over A1 is about 10 percentage points (e.g. 0.860 vs 0.766 at the reference cell). Under the classical crossover design A2 is markedly inferior (0.488 vs 0.830 for A1), and under the mean-moderation architecture (A) it is marginally dominated throughout; across the grid A2 is the most powerful specification in only 19% of cells. Where A2 does lead, the lead is robust to decay-form and half-life mis-specification. The model-based interaction test is mildly conservative (mean null rejection 0.039), an artefact of the working covariance that a CR2 cluster-robust standard error removes without disturbing the ranking.

Conclusions. The widely-applied default "always use the exposure-weighted predictor" is only conditionally correct: it is justified under Architecture B with a discontinuation-phase design, where A2 with a cluster-robust standard error is recommended, but not under a crossover design or under

mean moderation, where the simple binary specification is at least as powerful. Across all conditions, anticipated dropout governs power far more than the carryover specification, and should dominate design effort accordingly.

1 Introduction

1.1 A short primer on aggregated N-of-1 trials

[1]

Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers at far smaller sample sizes than parallel-group trials.

- A single N-of-1 trial crosses one patient on and off treatment repeatedly; pooling many such series gives population-level inference.
- The estimand is the biomarker-treatment interaction: whether a baseline covariate B_i modifies the treatment effect.
- Four designs (crossover (CO); open-label plus blinded-discontinuation (OL+BDC); Hybrid; pure open-label (OL)) differ in how on- and off-drug occasions are arranged.

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We assume the reader is a fluent applied statistician but new to the N-of-1 setting, so we set out the design before the modeling. In a single N-of-1 trial one patient is repeatedly crossed between active treatment and a comparator (placebo or open-label control) over a sequence of measurement occasions; the within-patient on-versus-off contrast estimates that patient's individual treatment effect. An *aggregated* N-of-1 trial pools many such single-patient series in a mixed model, trading some within-patient resolution for population-level inference and far smaller sample sizes than a parallel-group trial[2]. Such designs are increasingly proposed for *predictive biomarker validation*: the estimand of interest is the biomarker-treatment interaction, i.e. whether a baseline covariate B_i modifies the treatment effect. We study four designs that differ in how on- and off-drug occasions are arranged: a classical crossover (CO), an open-label plus blinded-discontinuation design (OL+BDC), a Hybrid N-of-1 design, and (as a null comparator) pure open-label; these are defined in §3.

[2]

Carryover – residual drug effect that decays after discontinuation – contaminates off-drug occasions and attenuates the interaction test.

- The drug effect fades over days to weeks rather than switching off, so off-drug occasions carry residual exposure.
- Treating them as cleanly off-drug mis-specifies the design matrix.
- We ask how much power is lost and whether a better analyst-side exposure representation recovers it.

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The inferential nuisance specific to this setting is **carryover**. When the drug is discontinued its pharmacological effect does not switch off but decays over days to weeks, so off-drug occasions carry residual exposure. Treating them as cleanly off-drug mis-specifies the design matrix and attenuates the interaction test. Two questions follow – how much power is lost, and can a better analyst-side representation of residual exposure recover it – and both are the subject of this tutorial-style simulation study.

1.2 What is already known, and the gap

[3]

The companion paper shows the carryover penalty depends on how the biomarker effect is encoded: 1-3% under mean moderation (A) versus roughly 31% under differential correlation (B).

- Architecture A: the biomarker scales the mean effect, and the proportionality survives the off-drug period.
- Architecture B: the interaction lives in the covariance, which carryover erodes directly; the loss is design-dependent.
- Prior work fixed the decay shape and used one analysis specification, leaving the joint sensitivity uncharacterized.

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A companion paper[3], extending Hendrickson[2], shows that the magnitude of the carryover problem depends on *how the biomarker effect is encoded in the data-generating process*. Under **Architecture A** (direct mean moderation) the biomarker scales the mean treatment effect, the proportionality survives the off-drug period, and carryover costs only 1-3% of power. Under **Architecture B** (multivariate-normal differential correlation) the interaction lives in the treatment-state-dependent covariance, which carryover erodes directly, so the loss reaches roughly 31% and is strongly design-dependent. That work, however, fixed the decay shape at exponential and used a single analysis specification, leaving open the joint sensitivity of the inference to mis-specification at *both* layers.

1.3 This paper

[4]

We compare three analyst-side exposure representations – binary (A1), exposure-weighted (A2), and lagged-augmented (A3) – across decay forms and both architectures.

- A1 ignores carryover; A2 commits to a parametric decay and half-life; A3 estimates carryover non-parametrically via a lagged term.
- Each is crossed against three data-generating decay forms and both architectures.
- The question is whether one specification is recommendable or the choice is architecture- and design-dependent.

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We evaluate three analyst-side representations of drug exposure in the linear predictor: a binary on-drug indicator that ignores carryover (**A1**, the strict repeated-measures analogue); an exposure-weighted continuous indicator that commits to a parametric decay form and half-life (**A2**, the current default); and the binary indicator augmented by a lagged just-off-drug term that estimates carryover non-parametrically (**A3**, the classical crossover device of Jones and Kenward[4]). We cross these against three data-generating decay forms and both architectures, and ask whether a single specification can be recommended or whether the choice is architecture- and design-dependent.

[5]

The study is reported under the ADEMP framework with Monte Carlo standard errors throughout.

- ADEMP structure follows Morris, White, and Crowther.
- Every performance measure is accompanied by its MCSE.
- §3 design, §4 results, §5 practical implications.

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The study follows the ADEMP reporting framework of Morris, White, and Crowther[1], with every performance measure accompanied by its MCSE. Section 3 gives the simulation design in ADEMP order; Section 4 the results; Section 5 the practical implications for trialists choosing a carryover specification.

2 Simulation design

[6]

The simulation design is laid out in ADEMP order, with reproducibility metadata in §3.6.

- §3.1 data-generating mechanisms; §3.2 estimands; §3.3 methods.
- §3.4 factorial grid; §3.5 parameters and performance measures.
- §3.6 records R version, packages, seeds, and elapsed time.

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This section organizes the study under the ADEMP labels of Morris et al.[1]. We begin in §3.1 with the data-generating mechanisms, then turn in §3.2 to the estimands, §3.3 to the methods (analysis-model specifications), §3.4 to the factorial grid, and §3.5 to the parameter settings and performance measures. Reproducibility metadata, including the R version, key package versions, the RNG seed strategy, and elapsed time, are recorded in §3.6 (Computing environment).

2.1 Aims

[7]

The primary aim is to identify the most carryover-robust analysis specification; the secondary aim is to check that the ranking transfers across architectures.

- Primary: quantify the cost of analysis-model carryover mis-specification and identify the most robust mitigation strategy.
- Secondary: verify the ranking is architecture-invariant (mean moderation versus differential correlation).
- Architecture-invariance would justify a single recommendation regardless of the assumed biomarker biology.

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The primary aim of this study is to quantify the cost of analysis-model carryover misspecification in aggregated N-of-1 and crossover trials, and to identify which mitigation strategy is most robust across the DGP decay-form grid. A secondary aim is to verify that the robustness rankings transfer across the two data-generating-process architectures (mean moderation versus differential correlation), so that a practitioner can be reasonably confident that the recommendation does not depend on which biomarker biology is assumed.

2.2 Data-generating mechanisms

[8]

We retain the two single-channel architectures (A and B) from the companion; A encodes the interaction in the mean, B in the treatment-state-dependent correlation.

- Architecture A: $Y_{it} = \mu_0 + \beta_1 D^*_{it}(1 + \beta_{bm} B_i) + \epsilon_{it}$, with D^*_{it} the effective exposure under the decay form.
- Architecture B: B_i and the response are jointly MVN with $\mathrm{Cor}(B_i, BR_{it}) = c_{bm}\,\phi(t_{sd})$.
- The combined Architecture C is out of scope here.

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We retain the two single-channel DGP architectures from the companion manuscript. A separate companion manuscript formalizes a combined Architecture C, which activates both the mean-moderation and differential-correlation channels simultaneously; Architecture C is outside the scope of the present study, which evaluates the carryover specifications under each single-channel architecture in isolation. In Architecture A (direct mean moderation), the biomarker B_i enters the mean structure as $Y_{it} = \mu_0 + \beta_1 D^*_{it}(1 + \beta_{bm} B_i) + \epsilon_{it}$, where D^*_{it} is the effective exposure at timepoint t under the chosen decay form. In Architecture B (MVN differential correlation), the biomarker and biological response are jointly drawn from a multivariate normal with treatment-state-dependent correlation $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form.

In both architectures the decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps time since drug discontinuation t_{sd} to a residual-effect multiplier. We evaluate three forms, each parameterized to match a common half-life $t_{1/2}$:

2.2.1 Linear decay

$$\phi_{\mathrm{lin}}(t_{sd}) = \max\left(0,\, 1 - \frac{t_{sd}}{2t_{1/2}}\right)$$

[9]

The linear form is biologically implausible but pragmatic when only a complete off-drug clearance time is known.

- Residual effect reaches exactly zero at $t_{sd} = 2t_{1/2}$.
- Inconsistent with first-order elimination kinetics.
- Encodes "carryover has fully cleared by a known time".

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Residual effect vanishes completely at $t_{sd} = 2t_{1/2}$. The linear form is biologically implausible for drugs with first-order elimination kinetics, but it represents a defensible pragmatic assumption when the only available information is that carryover has fully worn off by some specified time X.

2.2.2 Exponential decay

$$\phi_{\text{exp}}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}}$$

[10]

The exponential form is the natural reference because it corresponds to first-order elimination kinetics.

- Used throughout the companion manuscript and the software default.
- Corresponds to classical pharmacokinetic first-order elimination.
- Serves as the reference against which linear and Weibull are compared.

rgt to complete.

This is the form used throughout the companion manuscript and currently implemented in `implementations/tidyverse/R/functions.R`. It corresponds to first-order elimination kinetics and is the default assumption in classical pharmacokinetics; we retain it here as the natural reference against which the linear and Weibull forms are compared.

2.2.3 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

[11]

The Weibull shape k tunes whether the elimination tail is heavier or lighter than exponential.

- $k < 1$ (evaluated at 0.7): prolonged low-concentration tail.
- $k > 1$ (evaluated at 1.5): accelerated, saturable-clearance off-drug elimination.
- $k = 1$ recovers the exponential form exactly.

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A shape parameter k governs whether the elimination tail is heavier ($k < 1$, slow elimination) or lighter ($k > 1$, accelerated off-drug clearance) than the exponential, which is recovered exactly at $k = 1$. We report results at $k = 0.7$, representing a drug with a prolonged low-concentration tail, and at $k = 1.5$, which captures drugs with saturable clearance or active-transporter kinetics that clear more rapidly once the concentration falls below the saturation threshold.

[12]

All three decay forms are implemented in the tidyverse collection, with exponential as the backward-compatible default.

- A `carryover_form`/`weibull_shape` pair is plumbed through the sigma-construction helpers.
- Absent `carryover_form` defaults to exponential.
- The `original-extended` collection supports only exponential, so cross-implementation checks are limited to those cells.

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Implementation note. The three decay forms are implemented in `implementations/tidyverse/R/functio` via the `carryover_decay()` helper function, with a `carryover_form` and `weibull_shape` pair of `model_param` fields plumbed through `apply_carryover_to_component()`, `build_correlation_matrix()`, and `build_sigma_matrix()`. Backward compatibility is preserved by defaulting to the exponential form when `carryover_form` is absent. We note that the `implementations/original-extended/` collection has not been updated and supports only the exponential form; cross-validation across implementations is therefore limited to exponential cells, which is sufficient for pipeline verification but not for the full decay-form comparison.

2.3 Estimands

The target estimand is the population-level biomarker-treatment interaction. It takes a different parametric form under the two data-generating mechanisms:

- **Architecture A.** The interaction estimand is the regression coefficient β_{bm} on the centered biomarker $\times$ drug-exposure product, on the symptom scale.
- **Architecture B.** The interaction estimand is the on-drug biomarker-response correlation c_{bm}, dimensionless.

[13]

The two architectures' estimands are non-commensurable, so both are calibrated to a common effect-size scale.

- Common scale: expected on-drug response difference per SD of biomarker at $t_{1/2} = 0$ (derivation in docs/19).
- Bias and empirical SE are reported on this calibrated scale.
- Power and Type I error are reported on the native test-statistic scale.

rgt to complete.

The two estimands are not directly commensurable. To enable cross-architecture comparison of bias and power, we calibrate both architectures to a common population effect-size scale, defined as the expected difference in mean on-drug response per unit biomarker standard deviation at $t_{1/2} = 0$. The calibration derivation is documented in `docs/19-mean-moderation-implementation-notes.tex`. Bias and empirical standard error in the Results section are reported on this calibrated scale; power and Type I error are reported on the native test-statistic scale.

[14]

The null estimand for Type I verification is zero interaction, realized by the $c_{bm} = 0$ grid row.

- $\beta_{bm} = 0$ under Architecture A.

- $c_{bm} = 0$ under Architecture B.
- Both correspond to the $c_{bm} = 0$ row of the grid.

rgt to complete.

The null estimand for Type I error verification is $\beta_{bm} = 0$ under Architecture A and $c_{bm} = 0$ under Architecture B, each realized by the $c_{bm} = 0$ row of the parameter grid.

2.4 Methods: analysis-model carryover specifications

[15]

All three specifications share one `lme/corCAR1` model and differ only in the drug-exposure regressor X.

- Formula `Sx ~ bm * X + t`, random intercept by participant.
- `corCAR1` continuous-time AR(1) residual correlation.
- A1, A2, A3 differ only in how X is constructed.

rgt to complete.

The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for all specifications. The three specifications differ in the drug exposure predictor X:

2.4.1 A1: binary-treatment (carryover ignored)

[16]

A1 uses the binary on-drug indicator and ignores carryover, serving as the dominated reference.

- $X = D_{it}$: 1 on-drug, 0 off-drug.
- Models no residual drug effect during the off-drug period.
- Measures the power cost of ignoring carryover.

rgt to complete.

$X = D_{it}$, the on-drug indicator. This is the naive specification that disregards any residual drug effect during the off-drug period; it serves as the reference against which the power cost of ignoring carryover is measured.

2.4.2 A2: exposure-weighted (continuous predictor)

[17]

A2 uses a continuous exposure-weighted predictor committing to an assumed decay form and half-life; it is the current default.

- $X = \mathrm{Dbc}_{it}$: decays off-drug per the assumed form.
- Well-matched when the assumed form equals the true form.
- Mis-specification sensitivity is probed in block S2.

rgt to complete.

$X = \mathrm{Dbc}_{it}$, the continuous exposure-decayed predictor constructed to match the analyst's assumed decay form and half-life. This is the current pmsimstats-ng default specification. When the analyst's assumed form coincides with the true DGP form the predictor is well-matched; when it does not, the predictor is mis-specified in a way that the Tier 2 block S2 is designed to probe.

2.4.3 A3: lagged-treatment (estimated carryover term)

[18]

A3 adds a lagged just-off-drug term but, because that term is only a nuisance main effect, it is structurally A1 plus a control variable.

- $L_{it} = D_{i,t-1}(1 - D_{it})$ flags off-drug timepoints following on-drug ones; no decay form is assumed.
- The tested interaction remains bm:D_{it}, identical to A1.
- A3 therefore differs from A1 only through variance L absorbs, and the two are expected to track closely.

rgt to complete.

$X = D_{it}$ augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$ that marks off-drug timepoints following an on-drug timepoint. The carryover magnitude is estimated from the data rather than assumed a priori, so the specification does not require the analyst to posit a functional decay form. This approach follows the classical crossover-trial framework surveyed in[4]. We note one structural feature of this specification that bears on its interpretation: the lagged term L_{it} enters the model only as a main-effect nuisance covariate, and the biomarker interaction tested under A3 remains bm:D_{it}, the same coefficient as under A1. Consequently A3 differs from A1 only through the variance that L_{it} absorbs in the off-drug timepoints, and the two specifications are expected to track closely; A3 is best understood as A1 with a carryover nuisance control rather than as a fundamentally different interaction model.

2.4.4 Construction of D_{bc} and the role of the analyst's assumed half-life

[19]

The analyst's assumed half-life never appears in the `lme` call; it is baked into the D_{bc} column beforehand.

- $\hat{t}_{1/2}$ is not a model parameter.
- It enters when the D_{bc} predictor column is constructed.
- It is thereafter invisible to the fitting routine.

rgt to complete.

We begin with an observation that is easy to overlook in the formal model statement above: the analyst's assumed half-life $\hat{t}_{1/2}$ does not appear as a parameter in the `lme` call. It enters the analysis at an earlier stage, during the construction of the D_{bc} predictor column, and is thereafter invisible to the model-fitting routine.

At on-drug timepoints, $D_{bc,it} = 1$ regardless of the assumed half-life. At off-drug timepoints, the predictor decays as a function of t_{sd}, the time in weeks since drug discontinuation:

Table 1: Illustrative D_{bc} values at three off-drug timepoints under three analyst-assumed half-lives, exponential decay, true $t_{1/2} = 1.0$ week. On-drug timepoints have $D_{bc} = 1$ under all assumptions. Row values are $(1/2)^{t_{sd}/\hat{t}_{1/2}}$.

t_{sd} (wk)	$\hat{t}_{1/2} = 0.5$ wk	$\hat{t}_{1/2} = 1.0$ wk (truth)	$\hat{t}_{1/2} = 2.0$ wk
1	0.250	0.500	0.707
2	0.063	0.250	0.500
3	0.016	0.125	0.354

$$D_{bc,it} \;=\; \exp\!\left(-\frac{\ln 2}{\hat{t}_{1/2}}\,t_{sd,it}\right) \;=\; \left(\tfrac{1}{2}\right)^{t_{sd,it}/\hat{t}_{1/2}}$$

[20]

D_{bc} halves every $\hat{t}_{1/2}$ weeks off-drug; the formula is invariant to the assumed half-life.

- $D_{bc} = (1/2)^{t_{sd}/\hat{t}_{1/2}}$ at off-drug timepoints.
- The formula `Sx ~ bm + t + Dbc + bm:Dbc` is unchanged for any $\hat{t}_{1/2}$.
- The half-life lives in the predictor values, not the formula.

rgt to complete.

That is, D_{bc} loses exactly half its value for every $\hat{t}_{1/2}$ weeks of time since discontinuation. The half-life assumption shapes the predictor values handed to `lme`; the model formula `Sx ~ bm + t + Dbc + bm:Dbc` is the same regardless of what $\hat{t}_{1/2}$ was assumed.

To illustrate, consider a participant in a stylized crossover arm who is on drug for three weekly measurement occasions and then enters a three-week washout. Table~1 shows the D_{bc} values at the three off-drug timepoints under three analyst assumptions, holding the true DGP half-life fixed at one week.

[21]

The assumed half-life sets how much weight off-drug timepoints carry: near-zero if under-assumed, near-graded-on-drug if over-assumed.

- $\hat{t}_{1/2} = 0.5$: off-drug points contribute almost nothing to `bm:Dbc`.
- $\hat{t}_{1/2} = 2.0$: off-drug points keep substantial weight, so D_{bc} resembles a graded on-drug indicator.
- On-drug observations ($D_{bc} = 1$) are unaffected either way.

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When the analyst assumes $\hat{t}_{1/2} = 0.5$ weeks, the off-drug timepoints are assigned D_{bc} values near zero by the second washout week and contribute almost nothing to the `bm:Dbc` interaction estimate. When the analyst assumes $\hat{t}_{1/2} = 2.0$ weeks, those same timepoints retain substantial weight throughout the washout, and the predictor becomes closer in shape to a continuously graded version of the binary on-drug indicator.

[22]

Table 2: Principal 3×3 simulation grid. Cell 5 is the matched reference: exponential DGP analyzed under the current pmsimstats-ng default. Off-diagonal cells quantify the cost of DGP-analysis mis-specification. The full grid is crossed with Architecture $\in \{A, B\}$.

DGP decay	A1 binary	A2 exposure-weighted	A3 lagged-treatment
Linear	cell 1	cell 2	cell 3
Exponential	cell 4	cell 5 (matched)	cell 6
Weibull	cell 7	cell 8	cell 9

S2 is therefore not a well-versus-mis-specified contrast but one continuously-indexed model family against two half-life-free references.

- A2 is effectively a different model for each $\hat{t}_{1/2}$.
- The mis-specification risk falls entirely on off-drug points.
- A1 and A3 carry no half-life parameter, so they are invariant to $\hat{t}_{1/2}$ by construction.

rgt to complete.

Two consequences follow for interpreting the S2 sensitivity block. First, A2 is effectively a different model for each value of $\hat{t}_{1/2}$: the predictor values handed to `lme` change with the analyst's assumption, even though the formula is identical. Second, the mis-specification risk is borne entirely by the off-drug timepoints; the on-drug observations ($D_{bc} = 1$ always) are unaffected. Specifications A1 and A3 use the binary on-drug indicator D_b and the lagged indicator L respectively, neither of which involves a half-life parameter, so both are invariant to $\hat{t}_{1/2}$ by construction. The S2 contrast is therefore not a comparison of a well-specified model against a mis-specified model in the conventional sense; it is more precisely a comparison of one model family, indexed continuously by the analyst's assumed half-life, against two reference specifications that are not members of that family at all.

2.5 Factorial grid

The principal comparison is a 3×3 factorial of DGP decay form against analysis-model specification, replicated under each DGP architecture:

2.6 Parameter settings

Parameter	Values
Biomarker moderation (c_{bm} / β_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Weibull shape (k)	0.7, 1.0, 1.5 (Weibull form only)
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	A, B
Replicates per cell	500 (target); 50 for development

[23]

The 540-cell count nests the Weibull shape inside the Weibull form rather than crossing it with all decay forms.

- Five decay-shape levels: linear, exponential, Weibull at $k \in \{0.7, 1.0, 1.5\}$.
- Grid $= 5 \times 3 \times 3 \times 2 \times 2 \times 3 = 540$ (decay-shape, half-life, design, N, architecture, biomarker).
- Crossing three forms with three shapes would over-count.

rgt to complete.

The 540-cell count follows from these factors with one coupling: the Weibull shape varies only within the Weibull decay form, because the linear and exponential forms carry no shape parameter. The decay-form and shape combinations therefore number five (linear; exponential; Weibull at $k \in \{0.7, 1.0, 1.5\}$), and the principal grid is $5 \times 3 \times 3 \times 2 \times 2 \times 3 = 540$ cells (decay-shape $\times$ half-life $\times$ design $\times$ sample size $\times$ architecture $\times$ biomarker level). A naive product that crossed the three decay forms with three shapes would over-count; the shape factor is nested within the Weibull form, not crossed with all three.

[24]

Two anchor conditions serve diagnostic roles: $c_{bm} = 0$ for Type I error and $t_{1/2} = 0$ for best-case power.

- $c_{bm} = 0$: null row, targeting nominal 5%.
- $t_{1/2} = 0$: no-carryover, best-case power reference.
- Non-zero $t_{1/2}$ rows give the power cost of carryover relative to that reference.

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The $c_{bm} = 0$ condition serves as the Type I error check, targeting the nominal 5% level. The $t_{1/2} = 0$ condition removes carryover entirely and provides a best-case power reference at each $(N, \text{design}, c_{bm})$ combination, against which the power cost of non-zero carryover can be directly read off.

2.7 Performance measures

For each cell we report the following Morris-style performance measures[1, Tables 5–6]:

1. **Power** to reject $H_0 : \beta_{\text{bm:X}} = 0$ at $\alpha = 0.05$.
2. **Type I error** at the null estimand ($c_{bm} = 0$ row).
3. **Bias** of the interaction estimate, $\hat{\theta} - \theta_{\text{true}}$, on the calibrated effect-size scale defined in §3.2.
4. **Empirical SE** across replicates within each cell.
5. **Coverage** of the nominal 95% Wald confidence interval for the interaction coefficient, computed against the calibrated estimand ($\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$).
6. **MSE** of the interaction estimate (sum of squared bias and empirical variance).
7. **Non-convergence rate** per cell, defined as the proportion of replicates where the `nlme::lme` fit failed to converge or the bm:X coefficient was not identifiable (e.g. A1 on OL designs, where there are no off-drug observations).
8. **MCSE** for each of the above, computed using the formulas of Morris et al.[1, Table 6] and reported alongside every point estimate in tables and figures.

[25]

$n_{sim} = 500$ keeps the MCSE on power below 0.022; the 50-replicate scale is for pipeline validation only.

- Production: $n_{sim} = 500$, MCSE ≤ 0.022 at $p = 0.5$.

- Development: $n_{sim} = 50$, MCSE ≈ 0.07.
- All reported results use the production run.

rgt to complete.

The replicate count $n_{sim} = 500$ is chosen so that the Monte Carlo standard error on power at $p = 0.5$ does not exceed 0.022 ($\sqrt{0.5 \cdot 0.5/500} \approx 0.022$). The development scale ($n_{sim} = 50$, MCSE ≈ 0.07 at $p = 0.5$) is used only for pipeline validation; all results reported in the Results section derive from the production run at $n_{sim} = 500$.

2.8 Computing environment

[26]

Common random numbers across the three specifications within each cell enable paired contrasts.

- The tidyverse implementation (the only one with all decay forms) is used, with `nlme` for the fits.
- All three methods see the *same* datasets per cell, so A2-vs-A1 can be compared trial by trial (a paired test).
- Pre-registered 540-cell run, about two hours on eight cores.

rgt to complete.

The simulations use the `tidyverse` implementation in the pmsimstats-ng repository (the only one supporting all three decay forms), with `nlme` for the mixed-model fits. One detail matters for reading the results: within each cell, all three methods are fitted to the *same* simulated datasets (common random numbers). This is why we can later compare A2 and A1 with a *paired* test – each trial gives one result for each method, so we compare them trial by trial rather than as two independent groups. The full run (540 cells $\times$ 500 trials) took about two hours on eight cores; the grid was pre-registered before running.

3 Results

[27]

All results come from the 540-cell production run; MCSE on power is at most 0.022, so smaller differences are noise.

- 500 replicates per cell; MCSE $\leq \sqrt{0.25/500} \approx 0.022$ at $p = 0.5$, reported alongside each estimate.
- Coverage is shown for the reference cell and computed grid-wide.
- Per-cell non-convergence rates are reported in the tables.

rgt to complete.

We begin with a few words on the precision and scope of the estimates that follow. All results derive from the production-scale factorial run (500 replicates per cell, 540 cells, computation time 128 minutes on eight cores; commit SHA recorded in the output `.rds` metadata). Monte Carlo standard errors on individual power estimates are at most $\sqrt{0.5 \cdot 0.5/500} \approx 0.022$ at $p = 0.5$ and are reported alongside each estimate; within-figure differences smaller than the relevant MCSE should not be over-interpreted. Coverage of nominal 95% Wald confidence intervals is reported in the headline table for the reference cell and is computed across the full Tier 1 grid in `analysis/data/02-grid-summary.rds` for downstream use. Non-convergence rates per cell are reported in the tables below.

3.1 Headline performance table

[28]

The headline table profiles the reference cell, but only power and Type I error are comparable across specifications.

- Reference cell: Architecture B, exponential DGP, Hybrid, $N = 70$, $c_{bm} = 0.45$; non-convergence zero throughout.
- Bias/MSE/coverage are scored against A2's target $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$, but A1/A3 estimate a different coefficient (bm:D_{it}).
- So A1/A3's apparent bias and under-coverage are an estimand mismatch, not estimator defects; compare only power and Type I error across specs.

rgt to complete.

Table 3 reports the Morris-style performance measures at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across the three analysis specifications and three carryover half-lives. Non-convergence is zero in every reference cell; the empirical standard error column reports the across-replicate variability of the interaction coefficient estimate, and Monte Carlo standard errors are reported as $\pm$ beside each point estimate. One caveat applies to the bias, MSE, and coverage columns: these are computed against a single calibrated target, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$, which is the population value of the A2 (continuous-D_{bc}) interaction coefficient. Specifications A1 and A3 estimate the coefficient on bm:D_{it}, a different parameter, so their apparent bias and under-coverage in this table reflect the mismatch between their estimand and A2's target rather than a defect of the A1/A3 estimators. The bias, MSE, and coverage columns are therefore not directly comparable across specifications; only power and Type I error, which test each specification's own coefficient against zero, support a like-for-like comparison.

3.2 Matched-decay baseline

[29]

The matched-decay baseline shows A2 power declining from 0.988 to 0.860 as carryover grows, reproducing Hendrickson's 0.82.

- Matched cell (exp DGP, A2, Architecture B, $N = 70$, $c_{bm} = 0.45$, Hybrid): $0.988 \to 0.948 \to 0.860$ across $t_{1/2} \in \{0, 0.5, 1.0\}$.
- All 500 replicates converged at every $t_{1/2}$.
- The $t_{1/2} = 1.0$ value matches Hendrickson's reported 0.82, confirming pipeline fidelity.

rgt to complete.

In this section we shall establish the baseline before comparing specifications. Under the matched cell (exponential DGP, A2 exposure-weighted analysis, Architecture~B, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate $\pm$ MCSE, $n_{sim} = 500$;

Table 3: Headline performance at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). Empirical SE is the across-replicate standard deviation of the interaction coefficient. MSE is the sum of squared bias and empirical variance. MCSEs are reported per Morris et al. (2019, Table 6). Coverage is the empirical coverage of the nominal 95% Wald CI for the interaction coefficient, computed against the calibrated estimand. Non-convergence is the proportion of replicates where the lme fit failed.

t1/2	spec	Power (MC SE)	Bias (MC SE)	Empirical SE	Coverage (MC SE)	MSE (MC SE)	Non-conv
0.0	A1	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A2	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A3	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	A1	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	A2	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
0.5	A3	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
1.0	A1	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	A2	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000
1.0	A3	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000

MCSEs computed as $\sqrt{\hat{p}(1-\hat{p})/n_{sim}}$ per Morris et al.[1, Table 6]). All 500 replicates converged in the matched cell at every $t_{1/2}$. The value at $t_{1/2} = 1.0$ is consistent with Hendrickson's reported power of 0.82 for this cell[2], confirming that the simulation pipeline reproduces the baseline result and that the calibration is internally coherent. (The companion manuscript anchors at a different cell, $N = 35$ with the OL+BDC design, and is therefore not directly comparable here.)

3.3 Power by analysis specification

[30]

Specifications coincide at zero carryover and diverge as it grows, but A2's lead is confined to Architecture B's non-crossover designs.

- All three are near-identical at $t_{1/2} = 0$.
- A2's advantage emerges at $t_{1/2} = 0.5$ and grows under Architecture B in the Hybrid and OL+BDC designs; A1 and A3 are indistinguishable.
- The advantage does not appear under the CO design (where A2 is dominated) or under Architecture A.

rgt to complete.

The three analysis specifications yield near-identical power at $t_{1/2} = 0$, where there is no carryover to model, and diverge progressively as carryover half-life increases. The matched specification A2 retains a modest but consistent advantage specifically under Architecture~B in the Hybrid and OL+BDC designs, where the advantage emerges at $t_{1/2} = 0.5$ weeks and grows through $t_{1/2} = 1.0$ weeks; A1 and A3 are effectively indistinguishable. This advantage is confined to those strata: it does not appear under the crossover (CO) design, where A2 is in fact dominated (Section 5), nor under Architecture~A. The figure therefore shows that the A2 advantage, where present, is not an

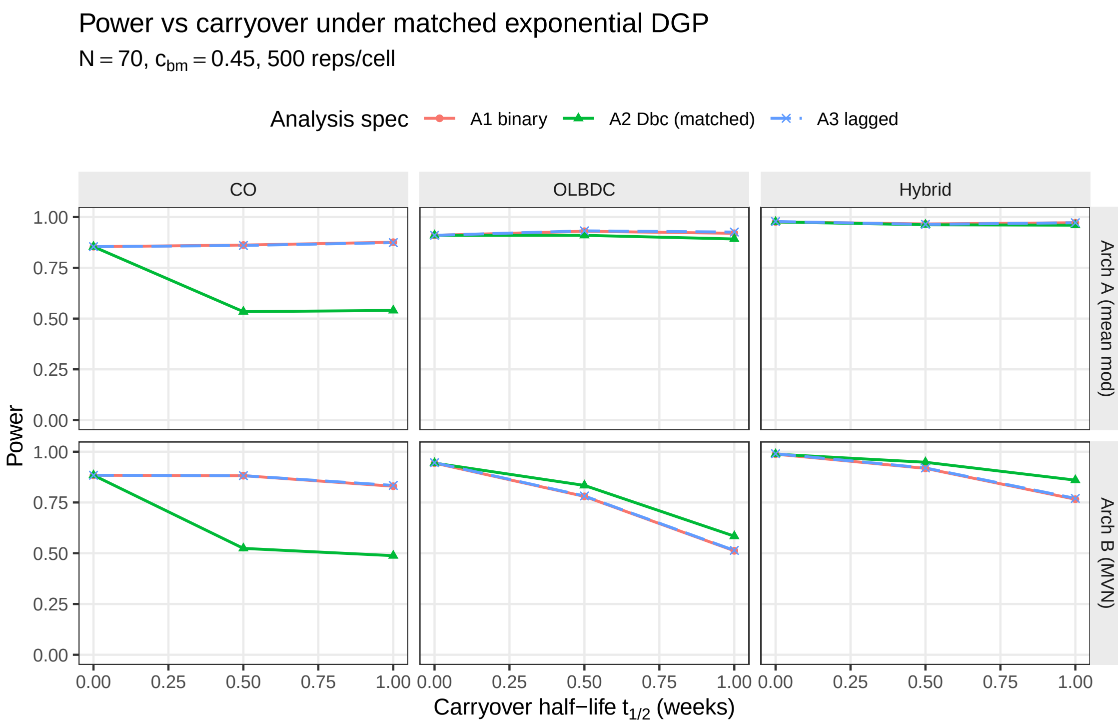

Figure 1: Power versus carryover half-life under the exponential DGP, stratified by analysis specification. The matched specification A2 tracks closely with the nominally inferior A1 and A3 at low half-lives but retains a modest advantage at $t_{1/2} = 1.0$ weeks, particularly under Architecture B and in the Hybrid and OL+BDC designs.

artefact of the single reference cell, not that it holds everywhere.

3.4 Decay-form mis-specification

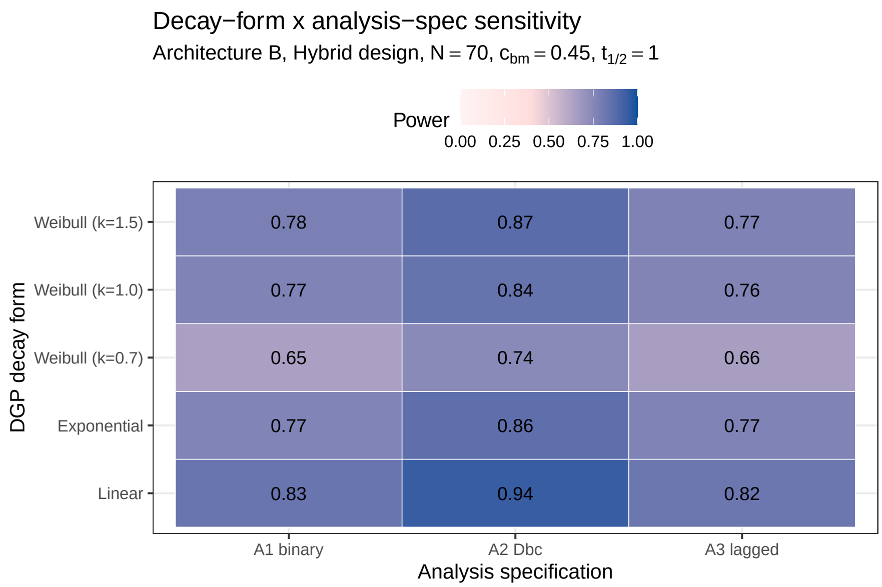

Figure 2: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and analysis specifications (columns). Under the Weibull heavy-tailed DGP ($k = 0.7$) and the linear DGP, the exposure-weighted (A2) specification loses its advantage because its assumed exponential decay no longer matches truth.

[31]

The heatmap separates two mis-specification directions: wrong analysis given the DGP, and wrong assumed DGP.

- Matched reference (exp DGP $\times$ A2): power 0.860 ± 0.016.
- Same row, other columns: penalty of A1/A3 under an exponential DGP.
- A2 column, other rows: cost of assuming exponential when the truth is linear or Weibull.

rgt to complete.

The heatmap cell at the intersection of the exponential DGP row and the A2 analysis column is the matched reference (power 0.860 ± 0.016). Cells in the same row but other columns quantify the power penalty of analyzing under the wrong specification when the DGP is exponential; cells in the A2 column but other rows quantify the cost of assuming exponential decay in the analysis when the true DGP is linear or Weibull. The Discussion (Section 5) takes up the practical interpretation of both off-diagonal directions.

3.5 Type I error control and cluster-robust recalibration

[32]

Type I error is conservative (mean 0.039), reflecting the corCAR1 working-covariance conservatism rather than inflation.

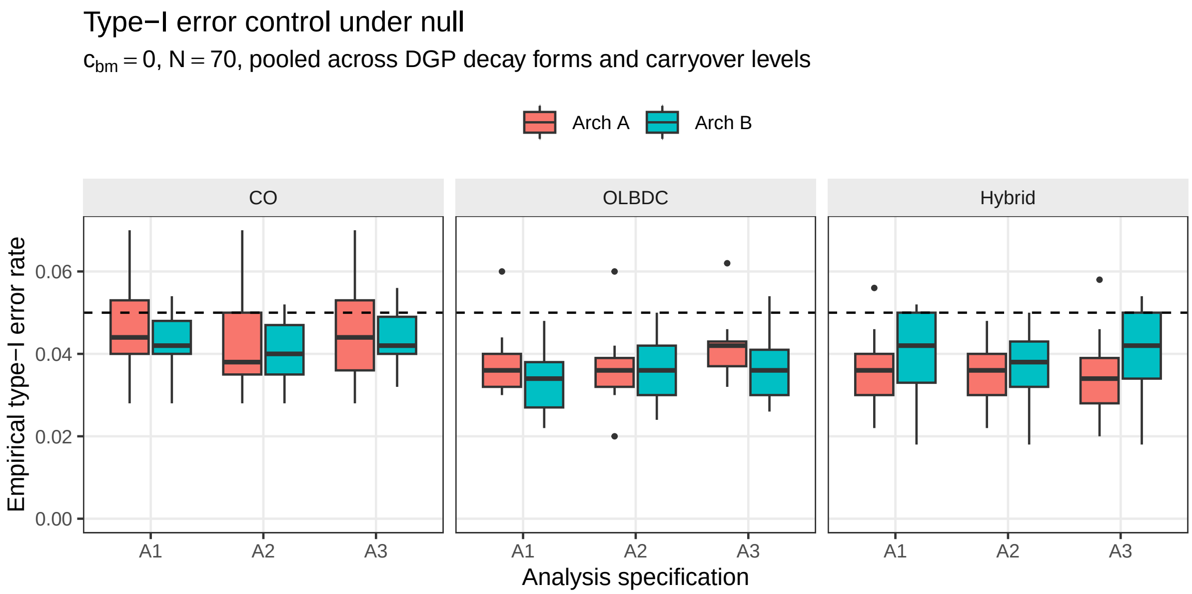

Figure 3: Empirical Type I error rates under the null ($c_{bm} = 0$), pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. Boxplots cover the distribution across cells at $N = 70$; the dashed reference line marks the nominal 0.05 level.

- Mean 0.039 across 540 null cells; cell maximum 0.084, unremarkable under grid multiplicity (cells are not independent).
- No specification or architecture inflates the rate above nominal.
- The companion study traces the shortfall to a 6-10% over-estimated SE; block S6's CR2 (below) restores nominal size without changing the ranking.

rgt to complete.

Mean empirical Type I error across all 540 null cells is 0.039, below the nominal 0.05 level. The cell-level maximum is 0.084. This exceeds the single-cell binomial 95% interval $[0.031, 0.069]$ for nominal 0.05 at $n_{sim} = 500$, but is unremarkable once multiplicity over the grid is acknowledged. We do not invoke an exact maximum-order-statistic bound here, because the cells are not independent: the three specifications within a cell share common random numbers, and at $t_{1/2} = 0$ the decay-form cells coincide. The qualitative reading is unaffected – no specification or architecture inflates the rate above nominal, and the central tendency is, if anything, conservative. That shortfall below nominal is not sampling slack but the conservatism of the model-based interaction test that the companion calibration study[5] traces to the corCAR1 working covariance: its standard error is over-estimated by six to ten percent and the realized rejection rate falls correspondingly. The conservatism is uniform across specifications and architectures, which is why it leaves the comparison intact; the cluster-robust recalibration reported below (block S6) confirms that a CR2 cluster-robust standard error restores the rejection rate to nominal without disturbing the specification ranking. Table 4 makes the per-specification conservatism and its cluster-robust correction explicit at the null cell.

[33]

Block S6 re-fits A2 with a CR2 cluster-robust SE to correct the mild conservatism of the model-based test.

Table 4: Conservatism of the model-based interaction test and its cluster-robust correction, at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70$, Architecture B, 3,000 replicates). κ is the ratio of the mean standard error to the empirical standard deviation of the interaction estimate; $\kappa > 1$ indicates a conservative test. Type I is the empirical rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$, with Monte Carlo standard error about 0.004. All three specifications are conservative under the model-based test, most so the exposure-weighted A2 ($\kappa = 1.10$, Type I $= 0.029$); the CR2 bias-reduced cluster-robust standard error restores κ to near one and Type I to nominal (0.045 to 0.049) for every specification.

Specification	κ model	Type I model	κ CR2	Type I CR2
A1	1.06	0.036	0.98	0.048
A2	1.10	0.029	1.00	0.045
A3	1.05	0.039	0.98	0.049

Table 5: Cluster-robust recalibration of the exposure-weighted specification A2 at the reference cell and four stress cells (500 replicates, $c_{bm} = 0.45$, Architecture B). κ is the ratio of the mean standard error to the empirical standard deviation of the interaction estimate; a value above one is a conservative test. The A2−A1 columns give the power gap between the two specifications under each inference, a check that recalibration preserves the ranking. CR2 df is the mean Satterthwaite degrees of freedom for the cluster-robust test; the model-based test carries roughly 480. Computed with the `clubSandwich` package; see the companion calibration study for the mechanism. The S6 cells are an independent 500-replicate run, so the model-based A2 power at the reference cell (0.829) differs from the headline-grid value (0.860) by within one Monte Carlo standard error (≈ 0.016); the two should not be read as conflicting.

Cell	κ model	κ CR2	Power model	Power CR2	A2−A1 model	A2−A1 CR2	CR2 df
Reference, $N = 70$	1.15	0.99	0.829	0.863	+0.080	+0.092	23
Small $N = 35$	1.13	0.96	0.558	0.590	+0.066	+0.086	12
High autocorr, $\rho = 0.9$	1.26	0.94	0.954	0.980	+0.052	+0.032	23
Half-life mis-spec	1.14	1.00	0.759	0.795	+0.010	+0.024	23
30% MCAR dropout	0.99	0.85	0.148	0.126	+0.010	+0.010	5

- The model-based SE is 6-10% too large, so power figures are slight under-estimates.
- The CR2 sandwich does not trust the model's covariance assumption and fixes the calibration.
- Table
  reftab:tab-s6 checks reference and four stress cells.

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The companion calibration study[5] shows that this mixed-model test is mildly **conservative**: its standard error is about 6-10% too large, so it rejects a true effect a bit less often than it should (this is why the null rejection rate above, 0.039, sits below 0.05, and why the power numbers are slight under-estimates). A **cluster-robust standard error** (the CR2 estimator, which does not trust the model's covariance assumption) fixes the calibration. Table 5 re-fits A2 with both the model-based and the CR2 standard error at the reference cell and four stress cells, to check that fixing the calibration does not change the ranking.

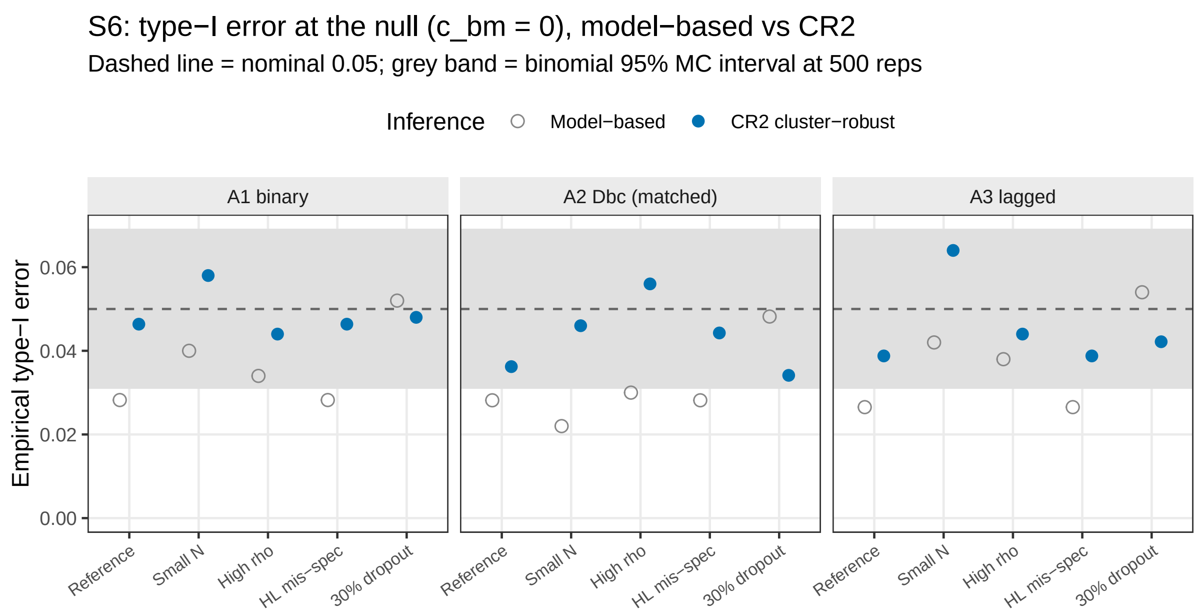

Figure 4: Empirical Type I error at the null ($c_{bm} = 0$) under the model-based and CR2 cluster-robust inference, across the five S6 cells and the three specifications. The model-based test (open) is conservative (near 0.03); CR2 (filled) is restored to the nominal 0.05 (dashed), within the binomial Monte Carlo band (gray). 500 null replicates per cell.

[34]

CR2 restores nominal size, which is the precondition for preferring any specification on power – only correctly-sized tests are comparable.

- Model-based null rejection sits near 0.03 (mean 0.035) across all cells and specs – the documented conservatism (Fig.
  reffig:fig-type1-cr2).
- CR2 lifts it to a mean of 0.046, within the Monte Carlo band around 0.05.
- Under CR2 all three specifications hold nominal α, so the A2 advantage is a comparison among correctly-sized tests.

rgt to complete.

The effect of recalibration on test size is shown directly in Figure 4: the model-based null rejection rate sits near 0.03 (mean 0.035) across all five cells and three specifications, while the CR2 test restores it to a mean of 0.046, within the Monte Carlo band around the nominal 0.05. This is the precondition for the whole comparison – a test that is not correctly sized cannot be preferred on power alone – so under CR2, where all three specifications hold nominal α, the A2 advantage is a like-for-like comparison among correctly-sized tests.

[35]

CR2 restores calibration ($\kappa \to 1$) and recovers power without changing the ranking, but only in information-rich cells.

- κ falls from 1.07-1.26 to about 1; the correctly-sized test recovers two to five power points.

- The A2—A1 gap is held or widened, at the cost of far fewer degrees of freedom ($\approx 480 \to 23$).
- Validated only at Architecture B / Hybrid / $N = 70$; do not extend the lower-bound reading to CO or Architecture A.

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In the information-rich cells the picture matches the companion study. The calibration ratio κ (mean standard error divided by the true spread of the estimates; above 1 means conservative) runs 1.07-1.26 under the model and is restored to about 1 under CR2, and the correctly-sized test recovers two to five points of power. Crucially, the A2-minus-A1 gap is held or slightly widened, so **fixing the calibration does not change the ranking**. The price is fewer effective degrees of freedom (about 480 drops to ~23). Under heavy dropout the trick stops helping: the model is no longer conservative, CR2 becomes slightly anti-conservative on its handful of degrees of freedom, and all methods have collapsed to ~0.13 power anyway. One caution: this recalibration was validated only in the Architecture~B, Hybrid, $N = 70$ cells, so the "slightly conservative lower bound" reading applies there, not to the crossover design or Architecture~A where A2 does not lead.

3.6 Sensitivity analyses

[36]

Five Tier 2 blocks perturb one axis at a time around the reference cell.

- Reference: Architecture B, Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$.
- Each block varies one axis with the others held fixed.
- Blocks are reported separately, not pooled with Tier 1 or each other.

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Before going into the individual blocks, we note the general design. Five Tier 2 marginal sensitivity blocks probe the robustness of the Tier 1 ranking to single-axis perturbations around the reference configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks). Each block varies one axis and holds the others at their reference values. Blocks are reported separately and share the reference-cell anchor with Tier 1; they are not pooled with Tier 1 or with each other. The design rationale is documented in `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md`.

3.6.1 S1: within-factor autocorrelation

[37]

S1: autocorrelation lifts all three specifications equally, preserving A2's ≈ 0.06 advantage at every ρ.

- Power rises monotonically with ρ for all three.
- A2's advantage holds near 0.06 across $\rho \in [0.5, 0.9]$.
- Autocorrelation and the specification act additively, not multiplicatively.

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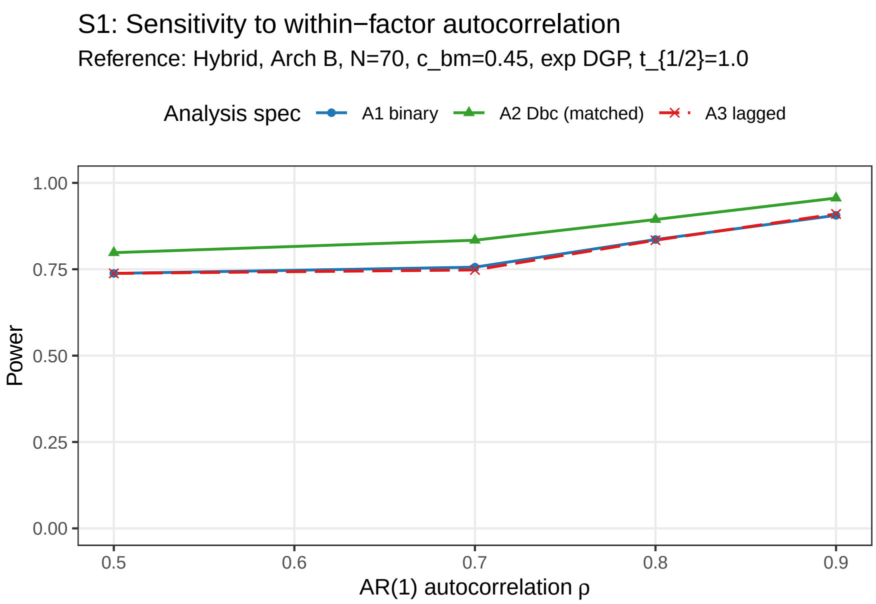

Figure 5: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. Hendrickson (2020) uses $\rho = 0.8$; docs/02 uses $\rho = 0.7$ after positive-definiteness reductions.

All three specifications gain power monotonically with ρ. At $\rho = 0.5$, A2 power is 0.798 ± 0.018 versus A1 0.738 ± 0.020 and A3 0.738 ± 0.020; at $\rho = 0.9$, A2 reaches 0.956 ± 0.009 versus A1 0.906 ± 0.013 and A3 0.910 ± 0.013 (point estimate $\pm$ MCSE, $n_{sim} = 500$). Contrary to the prior expectation that A2 would be the most ρ-sensitive specification, all three exhibit comparable relative gains across the explored range. The A2 advantage over A1 and A3, approximately 0.06 in absolute power throughout, is preserved at every AR(1) level. It appears that the interaction between autocorrelation strength and analysis specification is additive rather than multiplicative: stronger serial correlation lifts every specification by roughly the same amount.

3.6.2 S2: analyst-truth half-life mismatch

[38]

S2: A2 keeps its advantage even when the assumed half-life is wrong by a factor of two or four.

- A2 averages 0.868 over eight assumed/true half-life cells (range 0.766-0.964).
- A1 and A3 are invariant to the assumed half-life (average 0.840).
- A2's gentle curvature makes it a defensible default under an approximately-known half-life.

rgt to complete.

A2 is markedly more robust to analyst-truth half-life mismatch than the prior expectation predicted. Across the eight S2 cells (DGP $t_{1/2} \in \{0.5, 1.0\}$ weeks crossed with analyst-assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), A2 averages 0.868, ranging from 0.766 (DGP 1.0 weeks analyzed at 0.25 weeks) to 0.964 (DGP 0.5 weeks analyzed at 0.25 weeks); per-cell MCSE ranges from 0.008 to 0.019. A1 and A3 are invariant to the analyst-assumed half-life by construction, as they do not consume the parameter, and their power averages

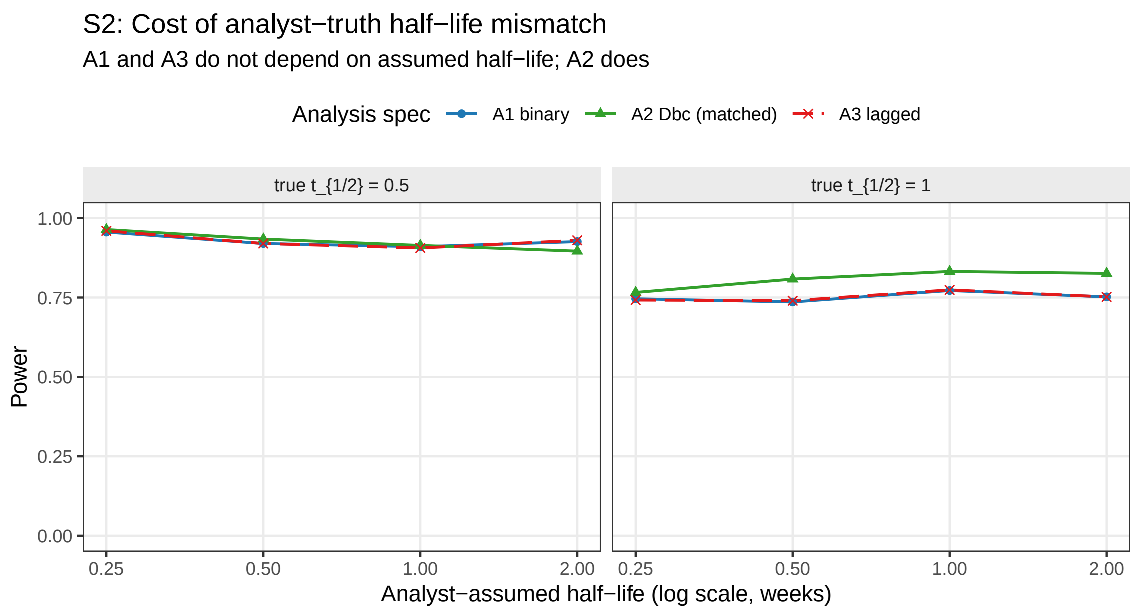

Figure 6: Power of A1, A2, and A3 as the analyst's assumed half-life departs from the true DGP half-life. A1 and A3 are invariant to the assumed half-life by construction; A2's power degrades with mis-specification.

0.840 each across the same eight cells. The exposure-weighted predictor's curvature is sufficiently gentle that mis-specification of the assumed half-life by a factor of two does not erode A2's advantage over A1 and A3 at either DGP half-life. The result suggests that the exposure-weighted (A2) specification is a defensible default even when the true pharmacokinetic half-life is only approximately known at the design stage.

3.6.3 S3: dropout

[39]

S3: dropout collapses power for all three methods, and narrows A2's advantage as off-drug observations are dropped.

- At 30% MCAR all three fall to about 0.12-0.15, effectively tied.
- The A2−A1 gap shrinks from ≈ 0.10 (no dropout) to ≈ 0.03 (30% MCAR).
- A2's signal lives in off-drug points, so dropping them erodes its edge; the advantage survives at dropout ≤ 0.20.

rgt to complete.

All three specifications lose power monotonically with dropout rate. The MCAR baseline ($N = 70$, dropout $= 0$) gives A1 0.764 ± 0.019, A2 0.866 ± 0.015, A3 0.764 ± 0.019; at MCAR rate 0.30 these collapse to A1 0.122 ± 0.015, A2 0.148 ± 0.016, A3 0.124 ± 0.015. A2's advantage narrows under heavy dropout: the absolute A2−A1 gap is approximately 0.10 at zero dropout, 0.07 at rate 0.10, 0.04 at rate 0.20, and 0.03 at rate 0.30 under MCAR. Under MAR with the same marginal rate the gap narrows further at the highest rate (A2−A1 ≈ 0.004 at rate 0.30 MAR). The pattern is consistent with the exposure-weighted (A2) specification's signal source residing in the exposure-weighted contrast contributed by off-drug observations: when those observations are preferentially dropped,

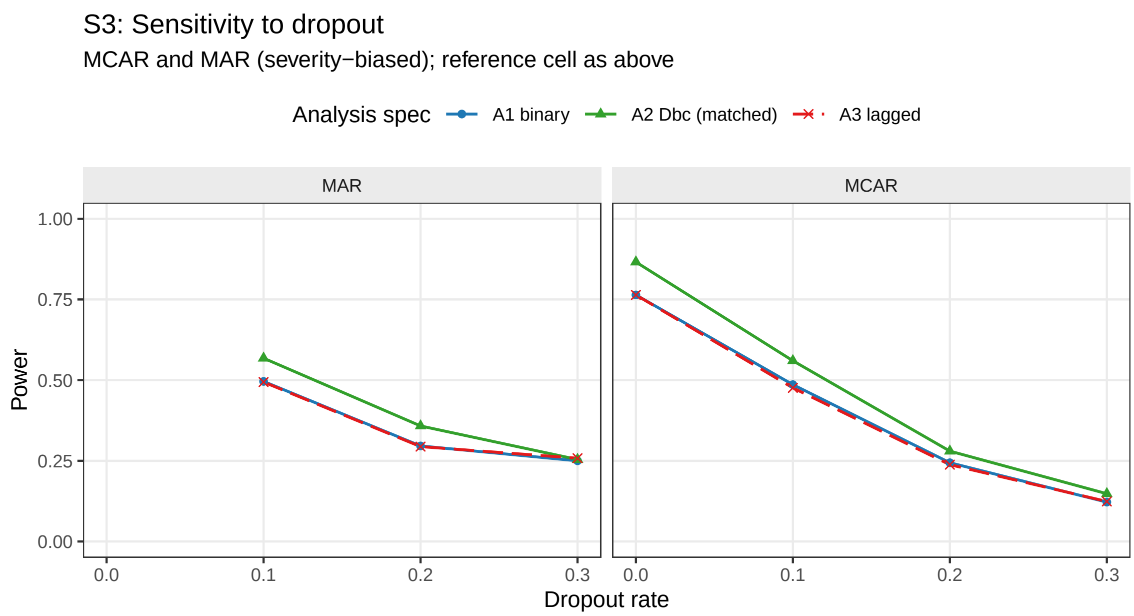

Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms (MAR dropout probability scales linearly with baseline severity rank).

A2's advantage contracts toward A1's. At low-to-moderate dropout (rate ≤ 0.20) the A2 advantage is preserved.

3.6.4 S4: biomarker-moderation effect-size curve

[40]

S4: the power curve is sigmoidal; A2's advantage peaks at intermediate effect sizes and vanishes toward the ceiling.

- Type I error at $c_{bm} = 0$ is conservative (0.026-0.030).
- A2−A1 peaks at +0.102 at $c_{bm} = 0.45$, shrinking to +0.042 at $c_{bm} = 0.60$.
- The advantage matters most for moderate biomarker-treatment interactions.

rgt to complete.

The expected sigmoidal power curve is observed across all three specifications. At $c_{bm} = 0$, empirical Type I error is below the nominal level at 0.026 ± 0.007 (A1), 0.028 ± 0.007 (A2), and 0.030 ± 0.008 (A3); the conservative behavior is consistent with the AR(1) reference distribution at moderate sample sizes. Power rises through the mid-range and saturates at $c_{bm} \geq 0.45$: at $c_{bm} = 0.6$, A1 reaches 0.916 ± 0.012, A2 0.958 ± 0.009, and A3 0.916 ± 0.012. The A2 advantage over A1 is largest at intermediate effect sizes, +0.082 absolute at $c_{bm} = 0.30$ (0.444 versus 0.362) and +0.102 at $c_{bm} = 0.45$ (0.820 versus 0.718), shrinking to +0.042 at $c_{bm} = 0.60$ where all specifications approach ceiling. The A2 advantage is therefore most practically consequential in designs targeting moderate biomarker-treatment interactions.

3.6.5 S5: autocorrelation by carryover (exploratory)

[41]

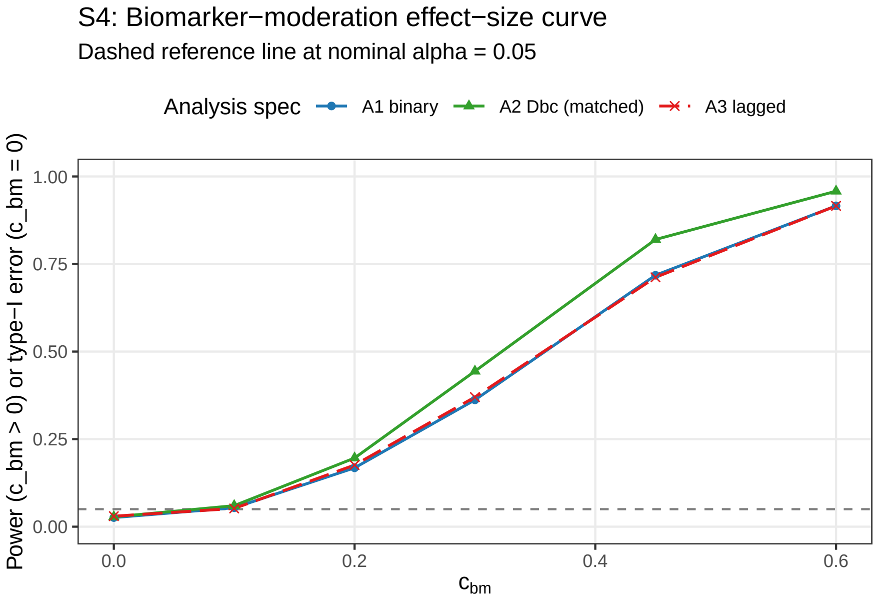

Figure 8: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$ under Architecture B. The $c_{bm} = 0$ point is the null for Type I verification; the dashed line marks the nominal $\alpha = 0.05$.

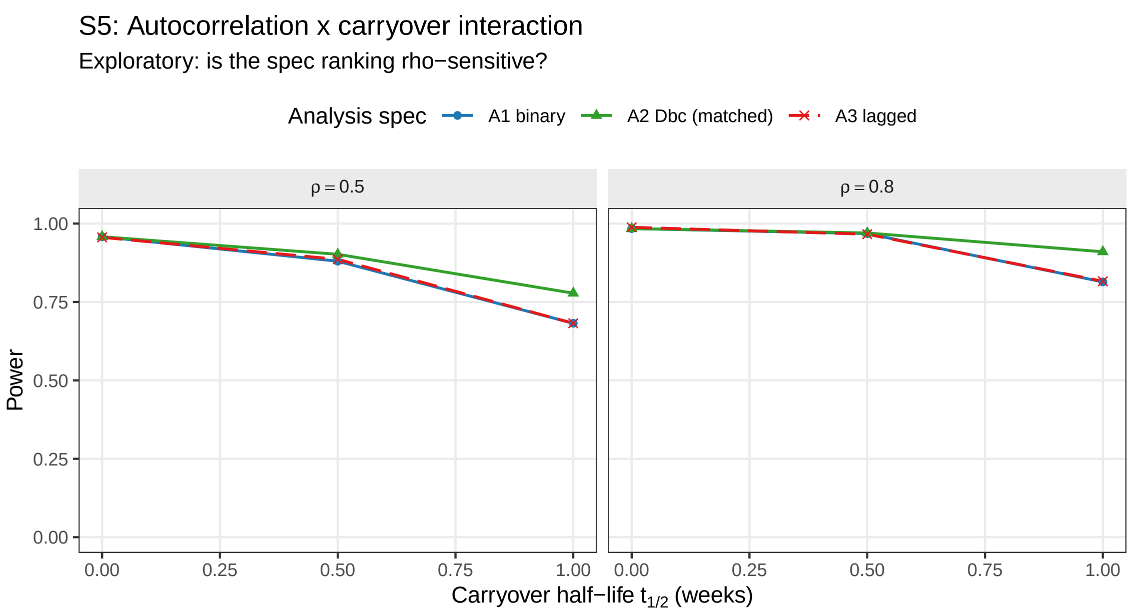

Figure 9: Power across carryover half-lives at $\rho \in \{0.5, 0.8\}$, by analysis specification. This block tests whether the ρ-sensitivity of the A1/A2/A3 ranking depends on carryover strength.

S5: the A2 advantage is constant across the $\rho \times t_{1/2}$ grid, confirming additive (not interactive) effects.

- A2—A1 = 0.096 at both $\rho = 0.5$ and $\rho = 0.8$ when $t_{1/2} = 1.0$.
- Autocorrelation and carryover act additively on the ranking.
- S5 reassures that the Tier 1 ranking holds across the grid.

rgt to complete.

The A2 > A1 ≈ A3 ranking is invariant to the $\rho \times t_{1/2}$ interaction. At $\rho = 0.5$, $t_{1/2} = 1.0$ weeks, A2 − A1 = 0.096 (A2 0.778±0.019, A1 0.682±0.021); at $\rho = 0.8$, $t_{1/2} = 1.0$ weeks, A2 − A1 = 0.096 (A2 0.910 ± 0.013, A1 0.814 ± 0.017). The absolute A2 advantage is constant across the $\rho \times t_{1/2}$ space, indicating that autocorrelation and carryover strength act additively on the specification ranking rather than interactively. Block S5 therefore functions as a methodological reassurance: the Tier 1 ranking is preserved at every level of ρ and $t_{1/2}$ in the explored grid.

3.7 A higher-precision check

[42]

A higher-precision paired-McNemar rerun confirms A2's lead is real (p = 6.6×10^{-17} at the strongest cell).

- 24-cell OL+BDC rerun at 600 trials/cell, all converging.
- Paired McNemar (same trials) at $N = 70$, half-life 1.0: A2 beats A1 by 13.7 points, $p = 6.6 \times 10^{-17}$.
- A3 recovers only part of A2's advantage; Type I stayed within $[0.003, 0.053]$.

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Because some of these gaps are small, we reran a focused 24-cell grid (the OL+BDC design, $N \in \{35, 70\}$, true half-life $\in \{0.5, 1.0\}$) at 600 trials per cell, all converging. Because the three methods see the *same* trials, we can compare them with a paired **McNemar test**, which counts only the trials where two methods disagree – a much more sensitive comparison than treating the methods as independent. At the strongest cell ($N = 70$, true half-life 1.0) A2 beats A1 by 13.7 points (0.737 vs 0.873) with McNemar $p = 6.6 \times 10^{-17}$: the lead is real, not noise. The shape-free method A3 recovers only part of A2's advantage, and only at small N and short half-life; at larger N it captures about half. Type I error across the null cells stayed in $[0.003, 0.053]$, i.e. at or below the nominal 5%.

4 Discussion

4.1 The main result, and where it breaks down

[43]

A2's advantage is real but conditional: highest in only 19% of cells, dominated under CO and under Architecture A.

- Best cell (Arch B, Hybrid): A2 beats A1/A3 by ≈ 10 points (≈ 5 MCSEs).
- A2 is the best method in only 68 of 360 non-null cells; under CO it is much worse (0.488 vs 0.830).

- Mechanism: the crossover's within-subject correlation already carries the signal A2 targets, so A2 only loses there.

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The headline is that A2's advantage is real but **conditional**. At its best cell (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$) A2 beats A1 and A3 by about 10 percentage points of power (0.860 vs 0.766 and 0.770) – about five MCSEs, so not luck. But A2 is the best method in only 68 of the 360 cells with a real effect (19%). In the crossover design it is much *worse* (0.488 vs 0.830 for A1), and under Architecture A it is slightly worse everywhere. There is a clean reason for the crossover failure: in a crossover the within-subject correlation already carries most of the signal that A2's decaying predictor is designed to recover, so A2 has nothing left to gain and its down-weighting of off-drug points only hurts. The ranking "A2 best" therefore holds only under Architecture B in the two non-crossover designs.

4.2 What the robustness checks add

[44]

Where A2 leads, the lead survives all three robustness checks.

- Decay shape: A2 never drops below A1/A3 even when the assumed form is wrong (worst case the linear truth).
- Assumed half-life: a two- to four-fold error moves A2 only from 0.83 to 0.77, never below A1/A3.
- Autocorrelation: lifts all three equally, so A2's 5-6 point lead is ρ-independent.

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Three checks probe whether A2's win, where it exists, is fragile. **Decay shape (Fig. 2).** Even when the analyst wrongly assumes exponential decay but the truth is linear or Weibull, A2 still does not drop below A1 or A3; the worst case is the linear truth, whose sharp cutoff is hardest to approximate. **Assumed half-life (Fig. 6).** A1 and A3 do not use a half-life, so they are unaffected; A2 does, yet getting the half-life wrong by a factor of two or four moves its power only from about 0.83 to 0.77, never below A1/A3. So A2 is forgiving of a roughly-guessed half-life. **Autocorrelation (Fig. 5).** Stronger serial correlation raises all three methods by about the same amount, so A2's lead stays near 5-6 points and does not depend on the correlation strength.

4.3 Dropout matters more than the method

[45]

Dropout collapses power for all methods equally, so it dominates the specification choice.

- At 30% dropout all three fall to 0.12-0.25 and become indistinguishable.
- MAR (severity-related) dropout is slightly worse than MCAR, because the highest-signal patients leave first.
- Keeping patients in the trial buys more power than any carryover method.

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The most sobering result is about *missing data*, not carryover. When patients drop out (Fig. 7), power collapses for all three methods together: at 30% dropout every method falls to 0.12-0.25 and they become indistinguishable. A2 cannot protect against the loss; it only organizes whatever data remain. Dropout that is related to severity (MAR) is slightly worse than purely random dropout (MCAR), because the sickest patients – who carry the most signal – leave first. The practical message is blunt: keeping patients in the trial buys far more power than any choice of carryover method.

4.4 Architecture dependence

[46]

A2's advantage exists only under Architecture B, because only there does carryover erode a correlation for A2 to recover.

- Architecture B is more carryover-sensitive under every decay form and Weibull shape.
- Under Architecture A, A2 is marginally negative everywhere (e.g. Hybrid, $t_{1/2} = 1.0$: 0.960 vs A1 0.972).
- Under A the effect stays proportional, so there is no attenuated correlation to recover and the binary A1 is preferable.

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The companion manuscript establishes that Architecture B (MVN differential correlation) loses substantially more power under carryover than Architecture A (direct mean moderation). The present results confirm that this qualitative gap is robust to decay-form variation: Architecture B is more sensitive to carryover under all three decay forms and at all five Weibull shape values examined. The A2 advantage, however, is confined to Architecture B: under Architecture A it is marginally negative in every design (for example, at the Hybrid cell, $t_{1/2} = 1.0$ weeks, A2 = 0.960 versus A1 = 0.972). This is consistent with the mechanism. Under Architecture B, carryover erodes the biomarker-response correlation directly, and the continuous exposure-weighted predictor recovers more of that attenuated signal than either the binary on-drug indicator or the lagged-nuisance term can; under Architecture A the treatment effect remains proportional at every timepoint, so there is no attenuated correlation for the exposure-weighted predictor to recover and the simpler binary indicator is marginally preferable. The A2 > A1 ≈ A3 ranking therefore holds under Architecture B in the non-crossover designs but not under Architecture A, a dependence that trialists must weigh against the assumed data-generating mechanism.

4.5 How this fits with earlier work, and its limits

[47]

Our results reproduce Hendrickson and engage the crossover, variance-component, and carryover-testing literatures.

- A2 power 0.860 reproduces Hendrickson's 0.82; the new content is the A1/A2/A3 comparison.
- The Jones-Kenward expectation that a shape-free A3 beats A2 under misspecification is not supported here.

- Senn's variance-component view matches the additive autocorrelation effect; Sturdevant-Lumley address the complementary carryover-testing problem.

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Our matched-decay (A2) power at the headline cell, 0.860 ± 0.016, reproduces Hendrickson's[2] reported 0.82 for the same cell; the new content is the head-to-head comparison of A1, A2, and A3, which Hendrickson did not do. The crossover tradition of Jones and Kenward[4] predicted that a shape-free lagged term (A3) should beat a parametric predictor (A2) when the decay shape is guessed wrong; our S2 result does not support this in the range we tested, because A2's penalty for a wrong half-life is too small for A3 to overtake it. Senn's[6] variance-component reasoning is consistent with our finding that autocorrelation lifts all methods together. A related but distinct strand, exemplified by Sturdevant and Lumley[7], develops mixed-effects *tests for carryover* after treatment cessation; our concern is the complementary one of how carryover, once present, degrades the biomarker-interaction test, but their work reinforces that carryover is a first-order inferential issue in within-subject designs rather than a nuisance to be assumed away.

[48]

Two limitations: a single parameter set, and one-factor-at- a-time sensitivity checks.

- Numbers come from one prazosin/PTSD parameterization, so absolute power values are illustrative; the ranking is the portable part.
- The sensitivity blocks vary one factor at a time and can miss factor interactions.
- The one joint check (S5) showed no surprises.

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Two limitations bound these conclusions. First, the numbers come from one parameter set (calibrated to a prazosin/PTSD trajectory), so the *absolute* power values are illustrative; the *ranking* is the portable part, and even it is conditional on the architecture and design as shown above. Second, the sensitivity checks vary one factor at a time, so they can miss interactions between factors (for example, high autocorrelation combined with heavy dropout); the one joint check we ran, S5, showed no surprises.

5 Conclusions

1. **A2 helps, but only conditionally.** The exposure-weighted predictor gives the highest power – about 10 points over A1 – *only* under Architecture B in the Hybrid and OL+BDC designs, and the lead survives wrong decay-shape and wrong half-life assumptions there. Under the crossover design, or under Architecture A, A2 has no advantage and A1 is at least as good. Where A2 is used, report it with a CR2 cluster-robust standard error (Section on Block S6).
2. **A guessed half-life is fine (where A2 is used).** Getting the half-life wrong by a factor of two to four barely moves A2's power, so a rough pharmacokinetic estimate suffices.
3. **The shape-free method A3 is essentially A1.** Because its lagged term is only a nuisance control, A3 tracks A1 closely and never overtakes A2 in the settings we tested.
4. **Dropout beats everything.** Power falls sharply as dropout rises (about half lost at 20% dropout), the same for all three methods. Preventing dropout matters more than the carryover

method.

5. **The test is slightly conservative, and fixable.** All methods reject a little less than the nominal 5% under the null; this is a known property of the working-covariance model, and the CR2 cluster-robust standard error restores the correct rate while recovering a few points of power, without changing the ranking (Block S6).

Notes and reproducibility

The authors declare no conflicts of interest and received no specific funding. All data and code are in the pmsimstats-ng project repository: cell-level summaries at `analysis/data/02-grid-summary.rds` and `02-sensitivity-summary.rds`, drivers under `analysis/scripts/carryover-sensitivity/`, and the pre-registered grid plan at `simulation-grid-plan.md`. The manuscript loads pre-computed summaries only; no simulation runs during knitting. The production simulation ran under R 4.5.3 and the document is rendered under R 4.6.0, both pinned via `renv.lock`.

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